Total No. of Questions : 4]

SEAT No. :

PC414

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[6359]-534

S.E. (Artificial Intelligence & Machine Learning Engg.) (Insem)

COMPUTER NETWORKS

(2019 Pattern) (Semester-III) (218543)

Time : 1 Hour] *[Max. Marks : 30*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Assume suitable data, if necessary.***

Q1) a) Define Signal. Discuss Any two Types of signal. With diagram. **[5]**

b) What is Multiplexing? Differentiate between FDM and WDM. **[5]**

c) Prove Shannon's Hartley theorem with respect to channel capacity. **[5]**

OR

Q2) a) Define Noise. Discuss the Following types of noise with the example **[5]**

i) Thermal Noise

ii) Induced Noise

b) Convert (10011100) binary data into digital wave using NRZ-L &NRZ-I Method. **[5]**

c) Explain Frequency modulation with a suitable diagram. **[5]**

P.T.O.

Q3) a) Draw a TCP/IP Model And explain the Responsibilities of the internet layer. **[5]**

b) Define addressing and explain logical and port addressing with an example. **[5]**

c) Discuss Server-client Network architecture with a diagram. **[5]**

OR

Q4) a) Discuss the following topology with a diagram. **[6]**

i) Mesh Topology

ii) Star Topology

iii) Bus Topology

b) Explain the functions of the following layer of the ISO/OSI model with a diagram. **[5]**

i) Transport Layer

ii) Presentation Layer

c) Write a short note on: **[4]**

i) Router

ii) Switch

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